

PART 2 • RESPONSE ESTIMATE

Name: _____ Team: _____ Scenario #: _____

Title: _____

My measured bags in 20 min: _____ R = bags $\times$ 3: _____

Planning rate S: _____ bags/team-hour; factor: 0.75 / 0.50

Reason for factor and route adjustment: _____

Mission and chosen approach

What must be accomplished? _____

Chosen structure or alternative: _____

Sketch with dimensions, layers, drainage, access and movement route:

Material estimate

Counting rule and calculation: _____

Bags N (round up): _____; contingency if chosen: _____

Fill N $\div$ 75: _____ yd³; quarter-yard delivery order: _____ yd³

Other materials and dimensions: _____

For a basin: chosen depth _____; ft³ _____; gallons _____;

approximate water load _____ lb; freeboard/overflow: _____

Compare team counts

One team = 4 workers. Four teams = 16 workers.

One-team ideal time N $\div$ S: _____ hours

Four-team ideal time N $\div$ (4 $\times$ S): _____ hours

These estimates assume enough tools, safe space, fill and movement capacity.

Adjusted team-hours N $\div$ S: _____

What could prevent four teams from being four times as fast?

STAFFING & EQUIPMENT TO MEET THE REQUEST

Name: _____ Scenario #: _____

Deadline and usable work time

Requested completion window: _____ hours
Delivery wait _____; setup/finish allowance _____ hours
Usable production time: _____ hours
Minimum teams = round up $[N \div (S \times \text{usable hours})]$: _____
Production workers = teams $\times$ 4: _____
Extra support workers and jobs: _____
Total people requested: _____
No usable time or unsafe mission? State the fallback: _____

Equipment and materials to request

Resource Quantity/capacity Arrival/source
Shovels: _____
Gloves and safety glasses: _____
Movement equipment: _____
Fill delivery equipment/service: _____
Tarps/liners/fabric/other: _____
Lights, radios, cones, first aid: _____

Delivery, movement and actual schedule

Fill delivered from/to and expected arrival: _____
Filled-bag route, trips and cycle time: _____

Operators and loading/placing workers: _____
How staffing avoids double-counting: _____
Schedule: task / start / finish / people and equipment

Final estimated elapsed time: _____ hours
Meets deadline? _____; limiting stage: _____

Worker support

Water stock plus reserve: _____ gal; cooler/refill plan: _____
Food needed / servings / timing: _____
Rest and rotation: _____
Weather protection and first-aid contact: _____
Changes needed if deadline cannot be met: _____

DECISION, INCIDENT REVISION & AFTER-ACTION REVIEW

Name: _____ Scenario #: _____

9. Is this practical?

Circle: Proceed as proposed / Proceed with changes / Reject primary bag plan

Explain using resources, time, safe site conditions and the mission:

What should emergency managers change? _____

What will you protect first, and what may remain unprotected?

Stop/relocate trigger and who makes the decision: _____

10. Incident update

Card #: _____ Update: _____

Which assumptions or resources changed? _____

New bag/fill/team/transport/time/support estimates as needed:

Revised priorities and request to incident lead: _____

Revised expected completion / deadline result: _____

What uncertainty remains? _____

11. Short after-action review

Which part of your initial plan failed first? _____

Which physical-trial observation helped your plan most? _____

Why isn't 'we need 600 sandbags' an operational plan?

One change you would make in a second 20-minute trial:

Optional extensions

Compare a shorter targeted barrier with moving equipment. Calculate how much one extra cart helps and when shovels become limiting. Test a 10% bag contingency. Design a fair second trial changing only one variable. Explain why two teams with the same bag count can finish a mission at different times.